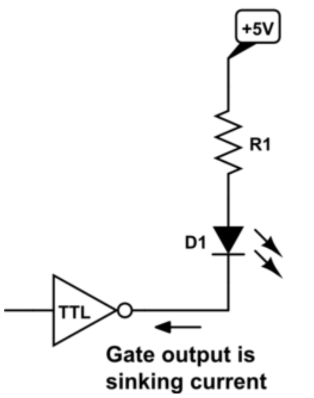
Switching Circuits Laboratory (CS29002)

Assignment 2

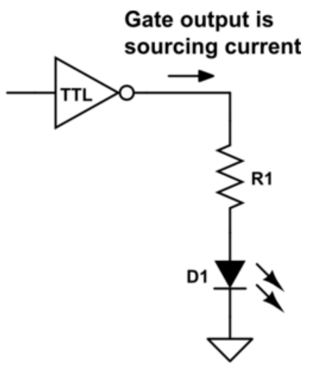
January 22, 2025

1

Interfacing a LED to a TTL output



LED glows when output is 0 (recommended)



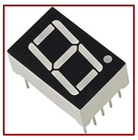
LED glows when output is 1 (not recommended)

A TTL gate output can
- source 400 microamperes (when 1)
- sink 16 (standard) of 8 (LS family) milliamperes of current.

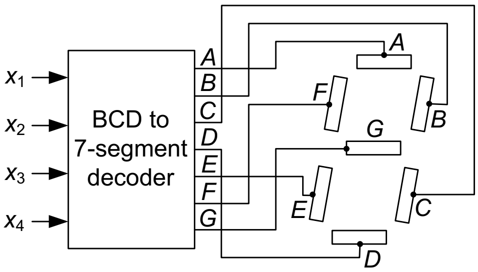
The voltage drop across a forward biased LED is typically 1.7 to 3.3 volts. For your calculation, assume it to be 2 volts.

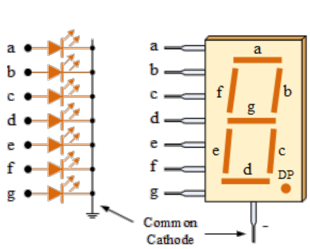
2

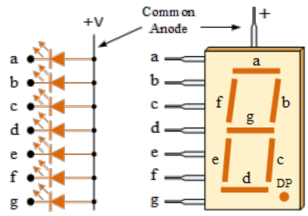
BCD to Seven-Segment Decoder



- 7-segment displays are commonly used for numerical displays.
  - There are seven display segments, designed using LED, LCD, or any other technology.
  - Depending on the digit to be displayed, the appropriate segments should glow.



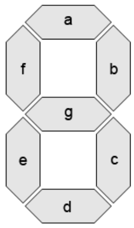




3

BCD Code **7-segment Code**

Digit	x_1	x_2	x_3	x_4	A	B	C	D	E	F	G
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	0	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	0	0	1	1



$A = x_1 + x_2'x_4' + x_2x_4 + x_3x_4$

$B = x_2' + x_3'x_4' + x_3x_4$

$C = x_2 + x_3' + x_4$

$D = x_2'x_4' + x_2'x_3 + x_3x_4' + x_2x_3'x_4$

$E = x_2'x_4' + x_3x_4'$

$F = x_1 + x_2x_3' + x_2x_4' + x_3'x_4'$

$G = x_1 + x_2'x_3 + x_2x_3' + x_3x_4'$

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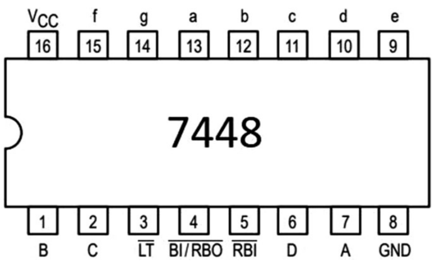
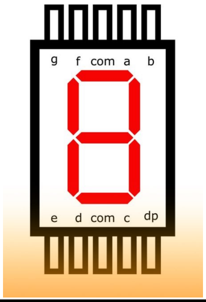


For common-anode display unit

For common-cathode display unit

Connect the LT and BI/RBO pins to +5V



Make sure to pair the display unit (common anode/cathode) with the proper decoder/driver IC

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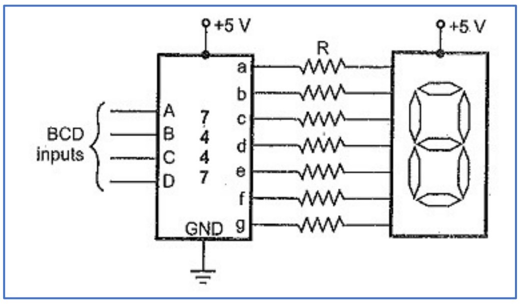
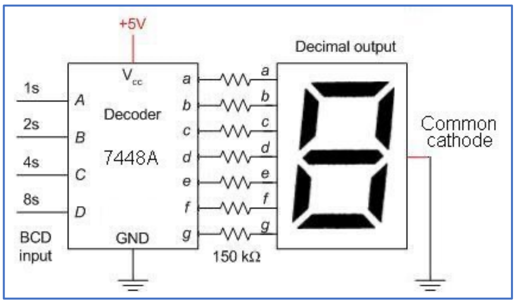


For common-anode display unit



For common-cathode display unit

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